Select Figures & Project Summary

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Single-Sequence Prediction of Andes Virus Evolutionary Constraints with ESM-2: Protein Language Models as a Rapid Surveillance Tool for Novel Outbreaks

Project Summary: This project aimed to explore whether protein language models (pLMs) could be used to predict how a virus may evolve even when sparse sequence data exists for that virus. Traditional evolutionary inference relies on multiple sequence alignments which may not be available in under-resourced areas. Andes virus is one such virus often affecting rural areas in Chile and Argentina. A multinational outbreak of a rare person-to-person outbreak of Andes Virus on the MV Hondius cruise ship, offered a test case opportunity.

The project used a pre-trained ESM-2 pLM to computationally mutate an inferred ancestral sequence and generate a grammaticality and semantic score for each amino mutation. The pLM embeddings were then used to investigate what predictions could be made regarding the outbreak sequence starting from the ancestral strain.

The results were mixed but provided broadly informative structural patterns and correctly but imperfectly flagged some known conserved sites. However, it was not fully reliable for individual residues. This indicates that pLMs show promise as surveillance tools in viral outbreaks and for pathogens that are under-sequenced. Andes virus may be an ideal pathogen to utilize the tool due to rare sporadic zoonoses and short transmission chains. The study also provides guidelines for future work.

Fig. 1

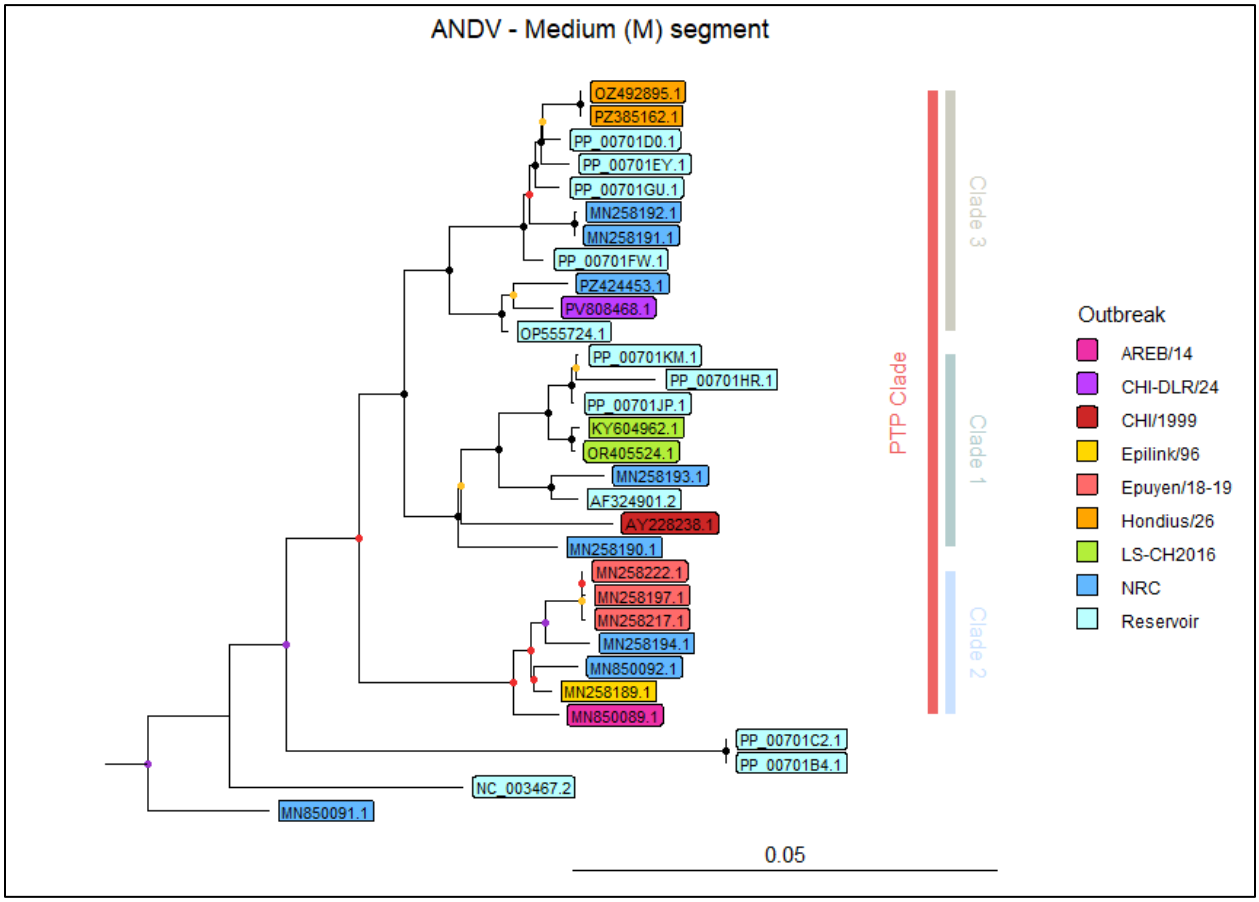

ANDV - Medium (M) segment
OZ492895.1
PZ385162.1
PP_00701D0.1
PP_00701EY.1
PP_00701GU.1
MN258192.1
MN258191.1
PP_00701FW.1
PZ424453.1
PV808468.1
OP555724.1
PP_00701KM.1
PP_00701HR.1
PP_00701JP.1
KY604962.1
OR405524.1
MN258193.1
AF324901.2
AY228238.1
MN258190.1
MN258222.1
MN258197.1
MN258217.1
MN258194.1
MN850092.1
MN258189.1
MN850089.1
PP_00701C2.1
PP_00701B4.1
NC_003467.2
MN850091.1
PTP Clade
Clade 3
Clade 1
Clade 2
Outbreak
AREB/14
CHI-DLR/24
CHI/1999
Epilink/96
Epuyen/18-19
Hondius/26
LS-CH2016
NRC
Reservoir
0.05

Fig. 2

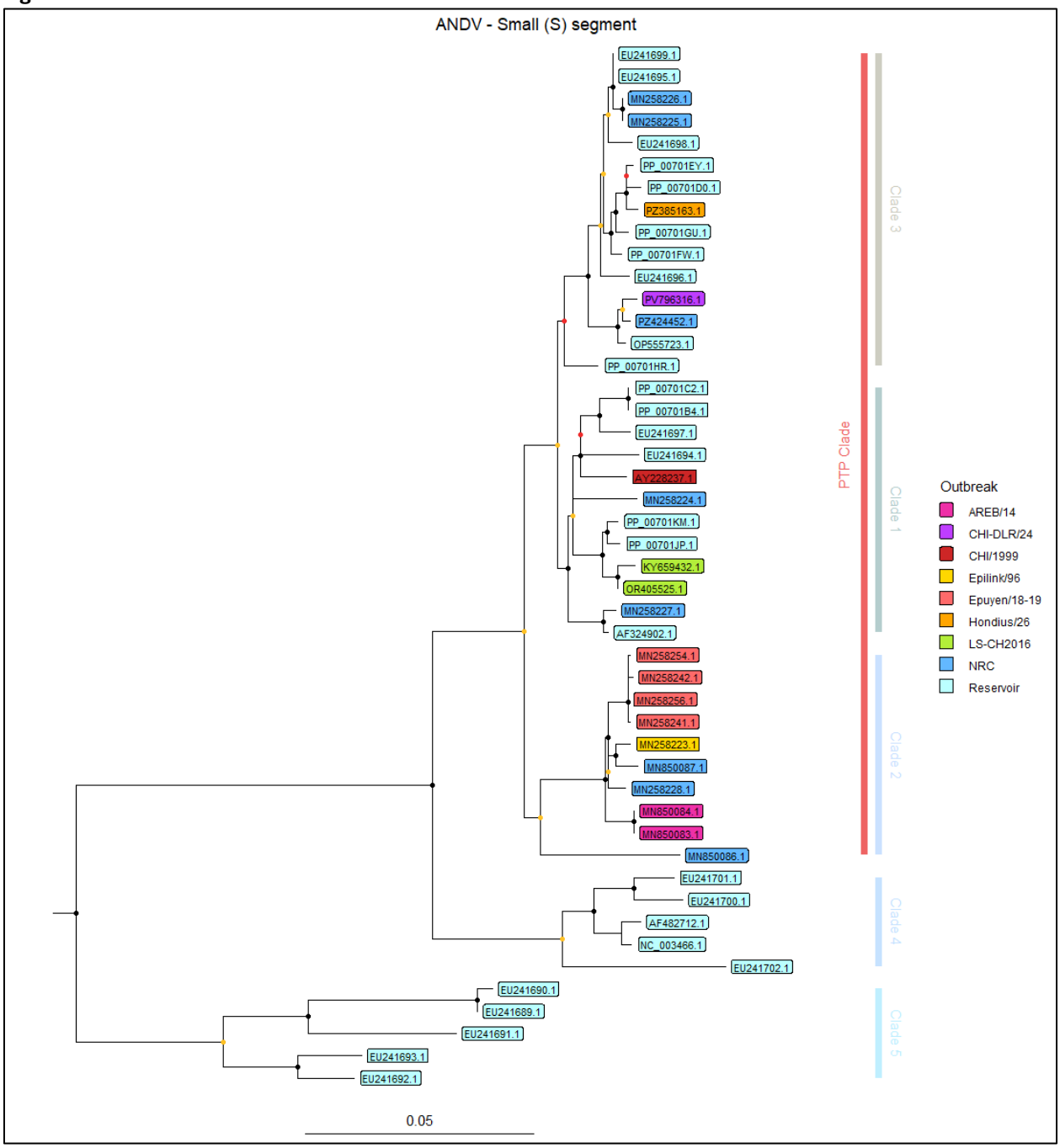


Figure 1& 2. M & S – Segment phylogeny of ANDV: Distinct clade differences between all known P2P outbreaks are visible within a hypothetical PTP clade. Other clades with only NRC or reservoir samples visible Bootstrap support is moderate – strong across most of the graph with colored nodes to show bootstrap support. Black = strong, yellow = moderate, red = weak, purple = below 50 bootstrap supports. Note small estimated nucleotide changes over branch length of 0.05. Clade naming adapted from Bellomo et al., 2023 and Medina et al., 2009)

Fig. 3

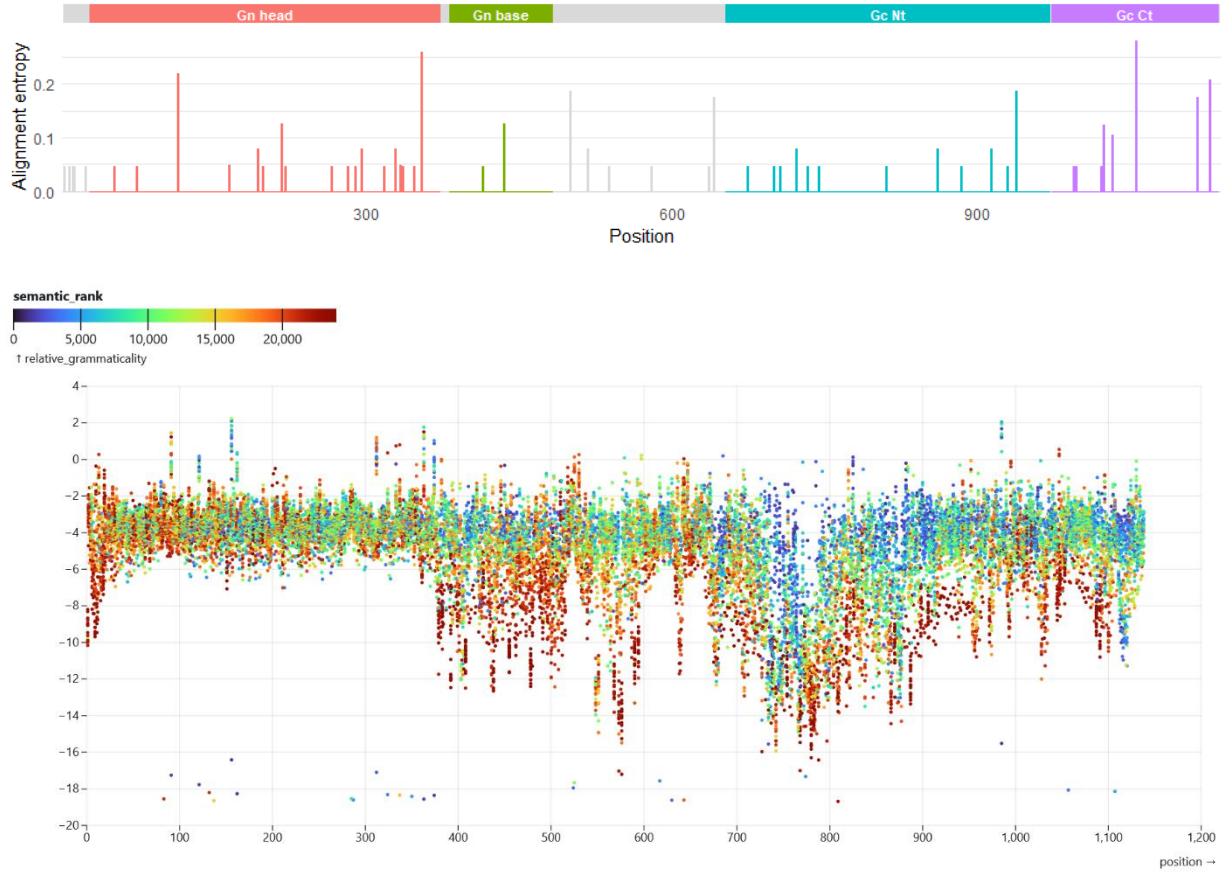


Fig 3 A/B. Glycoprotein DMS results: (top) MSA alignment per-site entropy graph with monomers defined, (bottom) **Relative Grammaticality** and **Semantic Score** plot. Note the semantic score has been converted in this graph to a rank to show greater differentiation of semantic changes. Lower relative grammaticality implies less chance of sequence existing. The X-axis 0 is the assumed null fitness (based on the exact sequence entered existing). Each point is calculated as a proxy of the likelihood of a sequence with only that point mutation existing. Semantic rank is a rank-transform of the semantic score which indicates the inferred change in structure. Thus, a much higher semantic score/rank implies the model is inferring that point mutation will yield greater change which is useful in determining molecular interactions. In-silico DMS and ESM-2 can thus calculate and be used to visualize the model's inference of all point mutations by their fitness and structural change. Regional changes or significant changes for either axis of different point mutations can signify important mutations to pay attention to in surveillance or describe evolutionary constraints across a region. Special thanks to Dr. Kieran Lamb who developed the javascript visualization environment on observablehq.com.